SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 8

# People Analytics: Social Network Analysis

People analytics usually uses individual HR records (role, team, pay, tenure). This week adds relationships through Social Network Analysis (SNA): employees = nodes, relationships = edges. Using a 360-degree feedback trial in a customer service department (34 nodes, 95 nominations), we cover nodes and edges (directed vs undirected), network-level measures (density, average path length, diameter, components), centrality (degree, in/out-degree, closeness, betweenness, eigenvector), the effect of removing nodes, and community detection with modularity and the Leiden algorithm — compared with formal teams. It ends with a Gephi Lite demo.

## Material

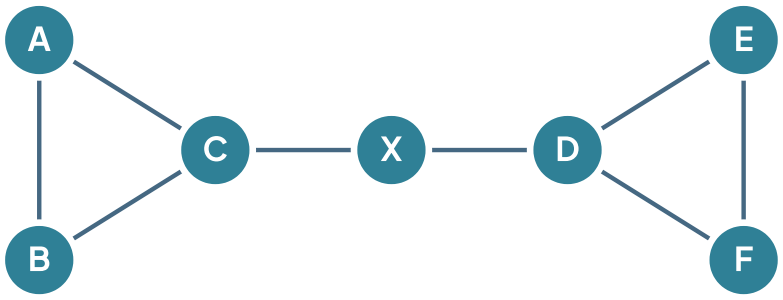
### 1 Session questions and place in the course

1. What is **social network analysis** and why use it in people analytics?
2. What do **density, average path length and modularity** measure?
3. How is the **centrality** of a node measured?
4. How are **communities** detected and how do they compare with teams?

In the course plan, People Analytics follows Marketing, Financial and Operations Analytics, before Big Data, Deep Learning, Generative AI, Data & AI Governance and the Capstone.

### 2 What SNA is and why it is used

**SNA** examines *relationships between individuals*, not only individuals, with measures at three levels: **network, node and community**.




| Source | Content |
|---|---|
| HR records | Describe individuals: role, team, pay, tenure. |
| Work | Happens through relationships (advice, feedback) that often cross reporting lines. |
| Questions | Who connects teams and which teams are silos need **relational data**. |

**Applications:** key people (who connects separate teams), new employees (who hasn't been nominated), teams & silos, change agents (whom to inform first), merged units (integrated yet?), collaboration load (who gets most requests).

**Org chart**

Formal reporting lines.

vs

**Network graph**

Working relationships: B reports to S1 but is connected to C and D in another team.

## 3 Nodes and edges: the 360-degree feedback trial

**Fictional data:** a customer service department ran a one-week 360-degree feedback trial; each employee nominated at least one colleague from any team or level. Participants: **27 department employees + 7 from finance, IT and sales**.

**Node list:** 34 nodes with role, team, year joined. **Edge list:** 95 nominations.

| Edge type | Meaning | Count |
|---|---|---|

| | | |
|---|---|---|
| **Directed** | From nominator to nominee | **95** |
| **Undirected** | Each pair once | **73** |

**4-row example**

Hendra→Sari, Sari→Hendra, Umar→Hendra, Vina→Umar = 4 directed, 3 undirected edges.

**Slide quiz**

Ani→Budi, Budi→Ani, Ani→Cici, Dedi→Cici, Cici→Dedi, Budi→Cici: **4 nodes, 6 directed, 4 undirected**.

## 4 Network-level measures: density, path length, diameter

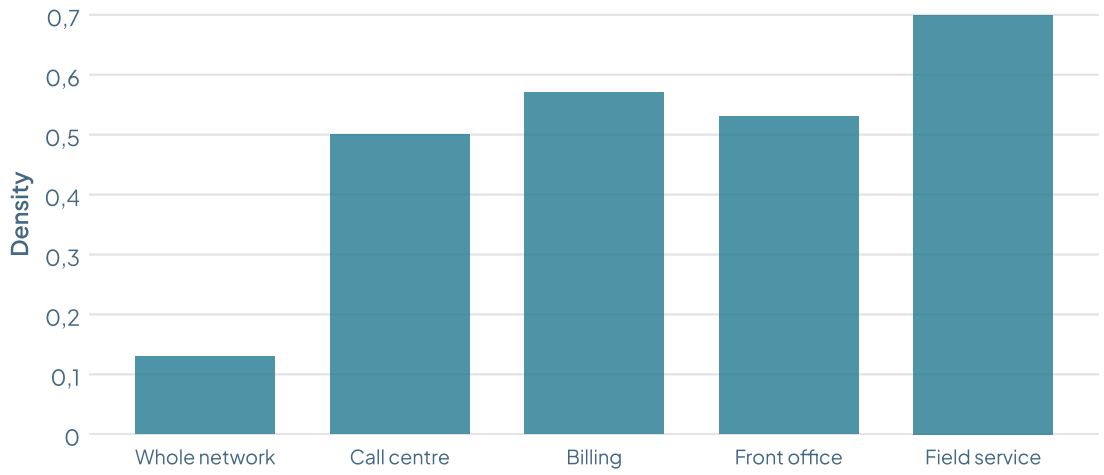
**Density** = share of possible edges that are present.

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Possible edges = n × (n - 1) ÷ 2 · Density = edges ÷ possible edges
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Example: 7 nodes → 21 possible; 8 present → **0.38**. (Quiz: 5 nodes, 6 edges → **0.60**.)

| Graph | Nodes | Edges | Possible | Density |
|---|---|---|---|---|
| Whole network | 34 | 73 | 561 | **0.13** |
| Call centre | 8 | 14 | 28 | 0.50 |
| Billing | 7 | 12 | 21 | 0.57 |
| Front office | 6 | 8 | 15 | 0.53 |
| Field service | 5 | 7 | 10 | **0.70** |

**Density: whole network vs each team**




Dense within teams, sparse across the network.

**Shortest path**: A→F via A, C, X, D, F = 4 edges. **Average path length**: mean shortest path over all pairs = 2.19 (example).

| Measure | Value | Meaning |
|---|---|---|
| Nodes | 34 | employees |
| Undirected edges | 73 | pairs with ≥1 nomination |
| Directed edges | 95 | nominations |
| Density | 0.13 | 73 of 561 |
| Components | 1 | all reachable |
| Avg path length | 2.71 | mean edges apart |
| Diameter | 6 | longest shortest path |

## 5 Centrality: degree, in-degree, out-degree

**Degree** = a node's number of edges; **degree centrality** = degree ÷ (n – 1). Node C: 3 ÷ 6 = 0.50.

### In-degree

### Out-degree

Incoming edges = nominations **received**.

Outgoing edges = nominations **made**.

**Slide quiz**

Highest in-degree: **Cici (3)**; highest out-degree: **Ani and Budi (2)**.

| Node | Role | Degree | Centrality | In | Out |
|---|---|---|---|---|---|
| Dewi | staff | 11 | 0.333 | 10 | 4 |
| Agus | supervisor | 10 | 0.303 | 9 | 4 |
| Sari | supervisor | 10 | 0.303 | 10 | 4 |
| Maya | supervisor | 9 | 0.273 | 8 | 4 |
| Joko | supervisor | 8 | 0.242 | 6 | 4 |
| Hendra | manager | 7 | 0.212 | 7 | 4 |

All six have the maximum out-degree 4; they differ in in-degree.

| In-degree 0 | Team | Joined | Out |
|---|---|---|---|
| Intan | call centre | 2026 | 1 |
| Kevin | call centre | 2026 | 1 |
| Zahra | front office | 2026 | 1 |
| Bayu | field service | 2026 | 1 |
| Iwan | field service | 2021 | 4 |
| Galih | sales | 2022 | 1 |

**HR insight**

Four of six joined in 2026 — an onboarding signal. Iwan nominated four but received none.

## 6 Closeness, betweenness, eigenvector

| Measure | Definition | Example | Organisational meaning |
|---|---|---|---|
| **Closeness** | Reciprocal of average shortest-path distance to all others | X: avg distance 1.67 → **0.60** | Fastest to reach everyone |
| **Betweenness** | Share of other pairs' shortest paths passing through the node | X on 9 of 15 pairs → **0.60** | Broker/bridge; a risk point if they leave |
| **Eigenvector** | High when neighbours are central | C, D highest **0.45** | Connected to influential people |

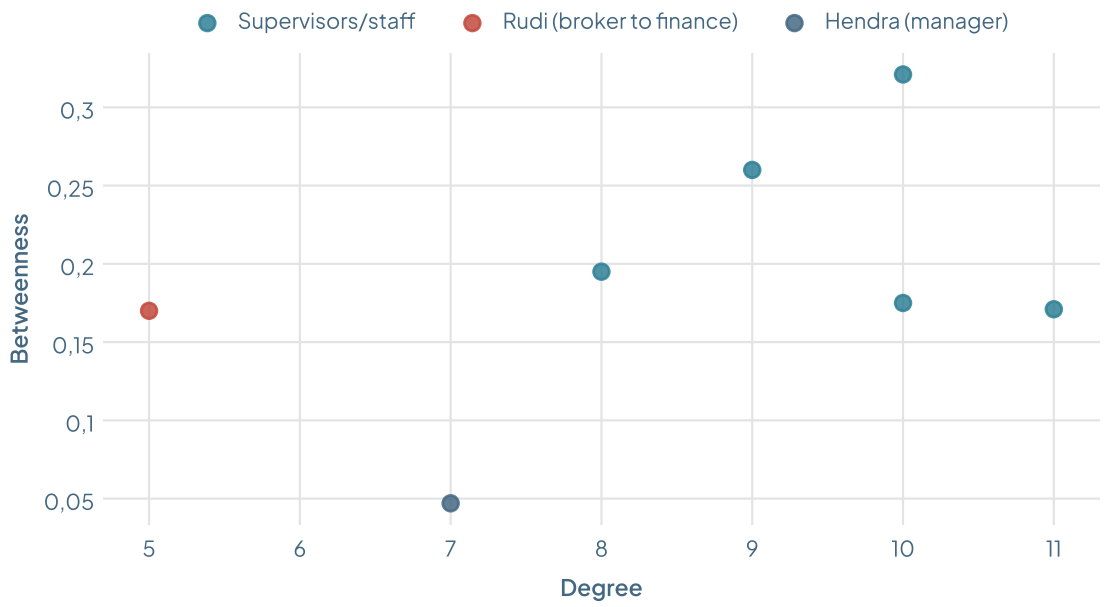
| Node | Degree | Closeness | Betweenness | Eigenvector |
|---|---|---|---|---|
| A, B, E, F | 0.33 | 0.40 | 0 | 0.33 |
| C, D | **0.50** | 0.55 | 0.53 | **0.45** |
| X | 0.33 | **0.60** | **0.60** | 0.38 |

**Most important node?**

It depends on the measure (slide quiz: P leads degree 0.50 & eigenvector 0.59; Q leads closeness 0.55 & betweenness 0.60).

| Measure | 1st | 2nd | 3rd | Hendra |
|---|---|---|---|---|
| Degree | Dewi 0.333 | Agus 0.303 | Sari 0.303 | 0.212, 6th |
| Closeness | Agus 0.541 | Sari 0.516 | Maya 0.516 | 0.493, 5th |
| Betweenness | Agus 0.321 | Maya 0.260 | Joko 0.195 | 0.047, 11th |
| Eigenvector | Sari 0.385 | Dewi 0.340 | Agus 0.336 | 0.327, 4th |

**Degree vs betweenness centrality (feedback network)**




Rudi (degree 5) and Dewi (degree 11) have nearly equal betweenness.

**Lesson**

High degree does not imply brokerage: Rudi (degree 5) and Dewi (degree 11) have nearly equal betweenness.

## 7 Removing a high-betweenness node

| Removed | Degree | Betweenness | Components | Cut off |
|---|---|---|---|---|
| Dewi | 11 | 0.171 | 2 | 1: Kevin |
| Sari | 10 | 0.175 | 2 | 1: Intan |
| Hendra | 7 | 0.047 | 1 | 0 |
| **Rudi** | 5 | 0.170 | 2 | **3: Ratna, Sinta, Teguh** |

Removing Rudi splits the network and cuts off three finance staff; removing the manager cuts off no one.

**Conclusion**

The effect of removing a node cannot be inferred from its degree.

## 8 Community detection and modularity

**Community detection** partitions nodes into groups with many internal and few external edges, using only **edges**, not team attributes.

| Aspect | Modularity |
|---|---|
| Definition | Share of edges within communities minus the share expected at random with the same degrees. |
| Reading | 0 = no structure; above ~**0.3** = clear structure. |
| Example | {A, B, C} and {X, D, E, F}: **0.37**. |
| Leiden | Algorithm finding the partition with the highest modularity. |

**Slide quiz**

Two communities {A, B, C, D} and {E, F, G} (modularity 0.36), linked by one edge D–E.

Leiden finds **five communities (0.52)** in the feedback network vs **0.44** for formal teams — communities describe structure better.

| Comm. | Call | Billing | Front | Field | Mgmt | Finance | IT | Sales |
|---|---|---|---|---|---|---|---|---|
| 1 | | | 6 | | 1 | | | 2 |
| 2 | 8 | | | | | | | |
| 3 | | | | 5 | | | 2 | |
| 4 | | 6 | | | | | | |
| 5 | | 1 | | | | 3 | | |

**Rudi**

In billing but assigned to community 5 with three finance nodes (3 of his 5 edges). Network data reveal the pattern; explaining it needs follow-up with

Rudi and Agus.

## 9 Density within and between teams: spotting silos

| | Call | Billing | Front | Field |
|---|---|---|---|---|
| Call centre | **0.50** | 0.07 | 0.06 | 0.05 |
| Billing | 0.07 | **0.57** | 0.02 | 0.03 |
| Front office | 0.06 | 0.02 | **0.53** | 0.03 |
| Field service | 0.05 | 0.03 | 0.03 | **0.70** |

**Conclusion**

Billing, front office and field service share one edge per team pair: they operate as **silos**, connected only through supervisors.

## 10 Demo: Gephi Lite in the browser

**Gephi Lite** runs in a browser (gephi.org/gephi-lite) with no installation or programming.

1. Open `feedback-directed.gexf`.
2. Size nodes by **in-degree**.
3. Find nodes with **in-degree 0**.
4. Size nodes by **total degree**.
5. **Detect communities**.

**Ethics (extra)**

Employee network data is sensitive: use it to improve collaboration and onboarding, not to punish individuals.

## Key takeaways

1. SNA studies relationships, not only individuals, at network, node and community levels. Node = employee, edge = relationship.
2. HR data describe individuals; work flows through relationships across reporting lines; silo/connector questions need relational data.
3. Directed edges run nominator → nominee (95); undirected join each pair once (73).
4. Density = edges ÷ [n(n−1)/2]; whole network 0.13, teams 0.50–0.70.
5. Shortest path, average path length (2.71), diameter (6), connected components (1).
6. Degree centrality = degree ÷ (n−1); in-degree = received, out-degree = made; in-degree 0 signals onboarding gaps.
7. Closeness = reciprocal of average distance; betweenness = share of shortest paths through a node (broker); eigenvector = connected to central nodes.
8. The most important node depends on the measure; high degree ≠ broker; removal effects can't be inferred from degree.
9. Community detection uses edges only; modularity above ~0.3 = clear structure; Leiden maximises it (0.52 vs 0.44 for teams).
10. Low between-team density = silos; network data reveal patterns, explanations need follow-up; Gephi Lite demo.

## Case studies

Case & worked answers

### Session case: reading the 360-degree feedback network (fictional slide data)

*Fictional company and people, as stated on the slides.* Three recommendations for Hendra: (1) onboarding buddies for the four 2026 joiners with in-degree 0; (2) broker risk — Rudi is the only bridge to finance, so build backup ties and include him in retention plans; (3) silos — cross-team customer-case forums for billing, front office and field service. To spread change fast, start with the highest-closeness nodes (Agus, Sari, Maya), not automatically the manager. Network findings always need confirming through conversation.

## Practice questions

Try answering first. The answer key is below each question.

**1. In SNA for people analytics, a 'node' and an 'edge' are…**

A. Team and department
B. An employee and a relationship between two employees
C. Role and pay
D. Manager and formal subordinate

**Answer: B.** Node = employee; edge = relationship.

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**2. Why is network analysis needed beyond ordinary HR records?**

A. HR records lack pay
B. Work flows through cross-line relationships; connector and silo questions need relational data
C. To replace the org chart
D. It's cheaper than surveys

**Answer: B.** It adds relationships to people analytics.

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**3. Ani→Budi, Budi→Ani, Ani→Cici, Dedi→Cici, Cici→Dedi, Budi→Cici. Number of UNDIRECTED edges?**

A. 3
B. 4
C. 5
D. 6

**Answer: B.** Unique pairs = 4 (6 directed).

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**4. A 7-node graph with 8 edges has density…**

A. 0.13
B. 0.38
C. 0.57
D. 1.14

**Answer: B.** 21 possible; 8 ÷ 21 = 0.38.

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**5. The feedback network has 34 nodes and 73 undirected edges. Density?**

A. 0.07
B. 0.13
C. 0.26
D. 0.50

**Answer: B.** 561 possible; 73 ÷ 561 = 0.13.

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**6. A network's diameter is…**

A. The average shortest path
B. The longest shortest path among all pairs
C. The number of components
D. The maximum number of edges

**Answer: B.** Feedback network: average 2.71, diameter 6.

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**7. In the directed 360 network, an employee's in-degree is…**

A. Nominations made
B. Nominations received
C. Team size
D. Tenure

**Answer: B.** In = received; out = made.

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**8. Four of six employees with in-degree 0 joined in 2026. The most sensible HR reading is…**

A. They perform poorly
B. New hires aren't connected yet — an onboarding signal
C. The data is wrong
D. They should be dismissed

**Answer: B.** Not nominated ≠ poor performance.

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**9. Dewi's degree centrality (degree 11) in a 34-node network is…**

A. 0.324
B. 0.333
C. 0.500
D. 11

**Answer: B.** 11 ÷ 33 = 0.333.

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**10. Which measure is the share of other pairs' shortest paths passing through a node (brokerage)?**

A. Degree
B. Closeness
C. Betweenness
D. Eigenvector

**Answer: C.** X lies on 9 of 15 pairs → 0.60.

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**11. A node scoring high because its neighbours are central is measured by…**

A. Eigenvector centrality
B. Out-degree
C. Density
D. Diameter

**Answer: A.** C and D score 0.45, linked to the triangle and X.

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**12. In the example, X has only 2 edges but the top closeness and betweenness. The lesson?**

A. X was miscounted
B. The most important node depends on the measure
C. Degree is always best
D. Closeness equals degree

**Answer: B.** C/D lead degree & eigenvector; X leads closeness & betweenness.

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**13. Removing Rudi (degree 5) cuts off 3 finance staff; removing Hendra (degree 7) cuts off none. Conclusion?**

A. Degree determines removal impact
B. Removal impact can't be inferred from degree
C. Managers are always most important
D. Finance doesn't matter

**Answer: B.** Rudi is a broker — key for retention and succession.

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**14. Community detection groups nodes based on…**

A. Formal team attribute
B. Edges only: many within, few between
C. Year joined
D. Role

**Answer: B.** It ignores team attributes, so it can be compared with teams.

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**15. Modularity 0.52 for Leiden vs 0.44 for formal teams means…**

A. Teams describe structure better
B. Leiden communities describe the structure better than formal teams
C. No community structure
D. The network split in two

**Answer: B.** Higher modularity = better partition; above ~0.3 is clear structure.

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**16. Between-team density among billing, front office and field service is only 0.02–0.03. This shows…**

A. Strong cross-team collaboration
B. The three teams work as silos, linked via supervisors
C. Teams are too big
D. Incomplete data

**Answer: B.** High diagonal, very low off-diagonal → silos.

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**17. Rudi (billing) is placed in the finance community. The analyst's best next step?**

A. Move Rudi to finance
B. Follow up with Rudi and Agus to understand why
C. Drop Rudi from the data
D. Conclude Rudi is disloyal

**Answer: B.** Network data reveal the pattern; explaining it needs follow-up.

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### Sources


- P3MD Week 8 MK6 — Business Analytics: People Analytics (Social Network Analysis)
- Gephi Lite (gephi.org/gephi-lite); Leiden algorithm (Traag et al., 2019)